

Ali Maatouk

Researcher on Foundation Models for Structured Data

📍 New York Metropolitan Area | ✉ ali.maatouk@yale.edu | 🔗 ali-maatouk.github.io
📄 Google Scholar: 2,346 citations | h-index: 22 | i10-index: 31

Summary

I am a researcher at the Yale Institute for Foundations of Data Science working on foundation models for structured data. My research focuses on developing multimodal and geometry-aware learning frameworks to enhance the representation, retrieval, and reasoning capabilities of foundation models across natural language, time-series, and structured data modalities. I have published over 50 papers in top venues across artificial intelligence, signal processing, and information theory, and have received multiple research awards for my work.

Professional Experience

Yale Institute for Foundations of Data Science

New Haven, CT

Researcher

Jan 2024 - Present

- Developed multimodal foundation models that jointly process time-series data and natural language to enable forecasting, question answering, retrieval, and reasoning over temporal data
- Created multimodal reinforcement learning frameworks that integrate time-series and natural language for temporal reasoning and explainable forecasting
- Designed and trained large-scale geometry-aware foundation models that dynamically learn and select the optimal geometric latent space for language modeling and retrieval
- Built graph-enhanced large language models by integrating contrastively trained graph neural networks to improve retrieval and multi-hop reasoning
- Designed agentic evaluation frameworks to automatically construct benchmark suites for time-series forecasting and text-grounded temporal reasoning

Huawei Technologies

Paris, France

Machine Learning Engineer

Sep 2021 - Dec 2023

- Designed and trained deep learning models to forecast network performance metrics and support proactive network optimization and anomaly detection
- Developed data processing and feature engineering pipelines for terabyte-scale network telemetry
- Leveraged graph-based learning approaches to model dependencies and interactions across distributed network systems and improve predictive performance
- Led research on large language models and retrieval-augmented generation for large-scale networked systems and developed the most cited benchmark dataset in the field
- Mentored research interns from leading institutions, including École Polytechnique and the University of Oxford, on projects involving machine learning and optimization

CentraleSupélec & University of Maryland, College Park

Paris, France & College Park, MD

Researcher — Joint PhD Program in Applied Mathematics

Oct 2017 - Aug 2021

- Introduced a new performance metric for quantifying information freshness in dynamical systems that has been widely adopted in the information theory community
- Developed mathematical frameworks based on stochastic hybrid systems and queueing theory to model information freshness in these dynamical systems under uncertainty
- Designed optimal control and resource allocation algorithms using Markov decision processes and restless multi-armed bandit formulations to optimize information freshness
- Derived performance bounds and structural properties for control policies in stochastic systems

Education

CentraleSupélec & University of Maryland, College Park

Paris, France & College Park, MD

PhD in Applied Mathematics

Oct 2017 - Dec 2020

Focus: Stochastic hybrid systems and Markov decision processes for maximization of information freshness in dynamical systems

CentraleSupélec

MS in Applied Mathematics | Ranked 1st in cohort

Focus: Information theory and signal processing

Lebanese University

BS in Computer Engineering | Ranked 1st in cohort

Paris, France

Sep 2016 - Sep 2017

Beirut, Lebanon

Sep 2012 - Aug 2016

Technical Skills

Programming Languages: Python, MATLAB

Deep Learning Frameworks: PyTorch, Hugging Face Transformers, DeepSpeed

Machine Learning Libraries: Scikit-learn, XGBoost, LightGBM, NumPy, SciPy, pandas

Awards and Honors

Oral Presentation at Learning on Graphs (LoG) Conference Dec 2025

ACM KDD Blue Sky Ideas Award for our work on foundation models for public health Aug 2025

Robert Innovation Fellowship from Yale University Feb 2025

AI Innovation Seed Funding from Yale University Nov 2024

Huawei Future Star Award for outstanding research contributions Dec 2021

Plateau de Saclay PhD Award for one of the best scientific contributions Dec 2020

André Blanc-Lapierre Award, National Prize for best Master's student in France in the domain of electricity, information technology, and communication Nov 2017

André Blanc-Lapierre Award, Trophée Ile-de-France for best Master's project in Paris region Nov 2017

Digicosme Scholarship for academic excellence Jul 2016

First Place in the Lebanese Baccalauréat official exams in Beirut Jul 2012

Publications

Google Scholar: 2,346 citations | h-index: 22 | i10-index: 31

2026

HypRAG: Hyperbolic Dense Retrieval for Retrieval-Augmented Generation

H. Madhu, N. Bui, [A. Maatouk](#), L. Tassiulas, S. Krishnaswamy, M. Yang, S. Ganguly, K. Srinivasan, R. Ying

International Conference on Machine Learning, 2026

TelecomTS: A Multi-Modal Observability Dataset for Time Series and Language Analysis

A. Feng, A. Varvarigos, I. Panitsas, D. Fernandez, J. Wei, Y. Guo, J. Chen, [A. Maatouk](#), L. Tassiulas, R. Ying

International Conference on Machine Learning, 2026

LitBench: A Graph-Centric Large Language Model Benchmarking Tool For Literature Tasks

A. Varvarigos, [A. Maatouk](#), J. Zhang, N. Bui, J. Chen, L. Tassiulas, R. Ying

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026

Fin-RATE: A Real-world Financial Analytics and Tracking Evaluation Benchmark for LLMs on SEC Filings

Y. Jiang, J. Chen, E. Makri, J. Chen, P. Li, [A. Maatouk](#), L. Tassiulas, E. Brenner, B. Xiang, R. Ying

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026

NumCache: KV Cache Compression and Retrieval for Financial Document QA

E. Makri, P. Li, Y. Jiang, J. Chen, J. Chen, [A. Maatouk](#), L. Tassiulas, E. Brenner, B. Xiang, R. Ying

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026

Age of Information Optimization in Distributed Sensor Networks with Half-Duplex Channels

P. Zou, [A. Maatouk](#), E. Erbayat, S. Subramaniam

IEEE International Symposium on Information Theory (ISIT), 2026

Parameter-Efficient Fine-Tuning of LLMs with Mixture of Space Experts

B. Zhang, J. Tao, Z. Zeng, N. He, [A. Maatouk](#), M. Yang, R. Ying
arXiv preprint arXiv:2602.14490, 2026

Multi-Modal Time Series Prediction via Mixture of Modulated Experts

L. Zhang, [A. Maatouk](#), J. Chen, L. Tassiulas, R. Ying
arXiv preprint arXiv:2601.21547, 2026

Reasoning through Verifiable Forecast Actions: Consistency-Grounded RL for Financial LLMs

J. Chen, A. Feng, H. Verma, S. Gu, H. Wang, [A. Maatouk](#), Y. He, Y. Gao, L. Tassiulas, R. Ying
arXiv preprint arXiv:2605.21975, 2026

2025

Position: Beyond Euclidean-Foundation Models Should Embrace Non-Euclidean Geometries

N. He, J. Liu, B. Zhang, N. Bui, [A. Maatouk](#), M. Yang, I. King, M. Weber, R. Ying
Learning on Graphs Conference (LoG), 2025 (Oral Presentation)

TRACE: Grounding Time Series in Context for Multimodal Embedding and Retrieval

J. Chen, Z. Zhao, G. Nurbek, A. Feng, [A. Maatouk](#), L. Tassiulas, Y. Gao, R. Ying
Conference on Neural Information Processing Systems (NeurIPS), 2025

HELM: Hyperbolic Large Language Models via Mixture-of-Curvature Experts

N. He, R. Anand, H. Madhu, [A. Maatouk](#), S. Krishnaswamy, L. Tassiulas, M. Yang, R. Ying
Conference on Neural Information Processing Systems (NeurIPS), 2025

LitFM: A retrieval augmented structure-aware foundation model for citation graphs

J. Zhang, [A. Maatouk](#), J. Chen, N. Bui, Q. Xie, L. Tassiulas, H. Xu, J. Shao, R. Ying
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025

MTBench: A Multimodal Time Series Benchmark for Temporal Reasoning and Question Answering

J. Chen, A. Feng, Z. Zhao, J. Garza, G. Nurbek, C. Qin, [A. Maatouk](#), L. Tassiulas, Y. Gao, R. Ying
arXiv preprint arXiv:2503.16858, 2025

Age of Information Optimization with Preemption Strategies for Correlated Systems

E. Erbayat, [A. Maatouk](#), P. Zou, S. Subramaniam
IEEE International Symposium on Information Theory (ISIT), 2025

Interdisciplinary development and fine-tuning of cardio, a large language model for cardiovascular health education in HIV care: Tutorial

R. Rullo, [A. Maatouk](#), T. Huang, J. Chen, W. Qiu, G. O'Connor, J. Womack, T. Sadak, C. Rodriguez, P. Carneiro, T. Espinosa, A. Marshall, R. Ying, S. Ramos
Journal of Medical Internet Research, Vol. 27, e77053, 2025

Telco-oRAG: Optimizing Retrieval-augmented Generation for Telecom Queries via Hybrid Retrieval and Neural Routing

A. L. Bornea, F. Ayed, A. De Domenico, N. Piovesan, T. S. Salem, [A. Maatouk](#)
IEEE Journal on Selected Areas in Communications, 2025

TeleQnA: A benchmark dataset to assess large language models telecommunications knowledge

[A. Maatouk](#), F. Ayed, N. Piovesan, A. De Domenico, M. Debbah, Z. Q. Luo
IEEE Network, 2025

Agoran: An Agentic Open Marketplace for 6G RAN Automation

I. Chatzistefan, N. Nikaein, A. Leone, [A. Maatouk](#), L. Tassiulas, R. Morabito, I. Pitsiorlas, M. Kountouris

Penalty Upon Decision: A Metric to Quantify Decision Costs in Status Update Systems

P. Zou, [A. Maatouk](#), J. Zhang, S. Subramaniam

IEEE Transactions on Networking, 2025

SANDWICH: Towards an offline, differentiable, fully-trainable wireless neural ray-tracing surrogate

Y. Jin, [A. Maatouk](#), S. Girdzijauskas, S. Xu, L. Tassiulas, R. Ying

IEEE International Conference on Machine Learning for Communication and Networking, 2025

Large-Scale AI in Telecom: Charting the Roadmap for Innovation, Scalability, and Enhanced Digital Experiences

A. Shahid, A. Kliks, A. Al-Tahmeesschi, A. Elbakary, A. Nikou, [A. Maatouk](#), et al.

arXiv preprint arXiv:2503.04184, 2025

2024

Pay Attention to What Matters

P. L. Silva, A. de Domenico, [A. Maatouk](#), F. Ayed

MINT@NeurIPS 2024, 2024

Tele-LLMs: A Series of Specialized Large Language Models for Telecommunications

[A. Maatouk](#), K. C. Ampudia, R. Ying, L. Tassiulas

arXiv preprint arXiv:2409.05314, 2024

Age of Information Optimization and State Error Analysis for Correlated Multi-Process Multi-Sensor Systems

E. Erbayat, [A. Maatouk](#), P. Zou, S. Subramaniam

MobiHoc, 2024

Large language models for telecom: Forthcoming impact on the industry

[A. Maatouk](#), N. Piovesan, F. Ayed, A. De Domenico, M. Debbah

IEEE Communications Magazine, 2024

Telco-RAG: Navigating the Challenges of Retrieval Augmented Language Models for Telecommunications

A. L. Bornea, F. Ayed, A. De Domenico, N. Piovesan, [A. Maatouk](#)

IEEE Global Communications Conference (GLOBECOM), 2024

Hermes: A Large Language Model Framework on the Journey to Autonomous Networks

F. Ayed, [A. Maatouk](#), N. Piovesan, A. De Domenico, M. Debbah, Z. Q. Luo

arXiv preprint arXiv:2411.06490, 2024

A framework for the evaluation of network reliability under periodic demand

[A. Maatouk](#), F. Ayed, S. Biao, W. Li, H. Bao, E. Zio

IEEE/ACM Transactions on Networking, Vol. 32(3), 2495-2510, 2024

2023

FlexTrain: A Dynamic Training Framework for Heterogeneous Devices Environments

M. Unsal, [A. Maatouk](#), A. De Domenico, N. Piovesan, F. Ayed

WANT@NeurIPS 2023, 2023

Age-aware stochastic hybrid systems: Stability, solutions, and applications

[A. Maatouk](#), M. Assaad, A. Ephremides

IEEE Journal on Selected Areas in Information Theory, Vol. 4, 762-783, 2023

An Optimization Framework For Anomaly Detection Scores Refinement With Side Information

A. Maatouk, F. Ayed, W. Li, Y. Wang, H. Zhu, J. Ye

IEEE Global Communications Conference (GLOBECOM), 6043-6049, 2023

How Costly Was That (In) Decision?

P. Zou, A. Maatouk, J. Zhang, S. Subramaniam

International Symposium on Modeling and Optimization in Mobile and Wireless Networks, 2023

2022

Analysis of an age-dependent stochastic hybrid system

A. Maatouk, M. Assaad, A. Ephremides

IEEE International Symposium on Information Theory (ISIT), 150-155, 2022

The age of incorrect information: An enabler of semantics-empowered communication

A. Maatouk, M. Assaad, A. Ephremides

IEEE Transactions on Wireless Communications, Vol. 22(4), 2621-2635, 2022

Semantics-empowered communications through the age of incorrect information

A. Maatouk, M. Assaad, A. Ephremides

IEEE International Conference on Communications (ICC), 3995-4000, 2022

Timely updates with priorities: Lexicographic age optimality

A. Maatouk, Y. Sun, A. Ephremides, M. Assaad

IEEE Transactions on Communications, Vol. 70(5), 3020-3033, 2022

2021

On the Global Optimality of Whittle's Index Policy for Minimizing the Age of Information

S. Kriouile, M. Assaad, A. Maatouk

IEEE Transactions on Information Theory, Vol. 68(1), 572-600, 2021

2020

Asymptotically optimal scheduling policy for minimizing the age of information

A. Maatouk, S. Kriouile, M. Assaad, A. Ephremides

IEEE International Symposium on Information Theory (ISIT), 1747-1752, 2020

The age of incorrect information: A new performance metric for status updates

A. Maatouk, S. Kriouile, M. Assaad, A. Ephremides

IEEE/ACM Transactions on Networking, Vol. 28(5), 2215-2228, 2020

On the age of information in a CSMA environment

A. Maatouk, M. Assaad, A. Ephremides

IEEE/ACM Transactions on Networking, Vol. 28(2), 818-831, 2020

On the optimality of the Whittle's index policy for minimizing the age of information

A. Maatouk, S. Kriouile, M. Assaad, A. Ephremides

IEEE Transactions on Wireless Communications, Vol. 20(2), 1263-1277, 2020

Status updates with priorities: Lexicographic optimality

A. Maatouk, Y. Sun, A. Ephremides, M. Assaad

WiOpt, 2020

2019

Age of information with prioritized streams: When to buffer preempted packets?

A. Maatouk, M. Assaad, A. Ephremides

IEEE International Symposium on Information Theory (ISIT), 325-329, 2019

Energy efficient and throughput optimal CSMA scheme

A. Maatouk, M. Assaad, A. Ephremides

IEEE/ACM Transactions on Networking, Vol. 27(1), 316-329, 2019

Minimizing the age of information in a CSMA environment

A. Maatouk, M. Assaad, A. Ephremides

WiOpt, 2019

Minimizing the age of information: NOMA or OMA?

A. Maatouk, M. Assaad, A. Ephremides

IEEE INFOCOM Conference on Computer Communications Workshops, 2019

2018

The age of updates in a simple relay network

A. Maatouk, M. Assaad, A. Ephremides

IEEE Information Theory Workshop (ITW), 1-5, 2018

On optimal scheduling for joint spatial division and multiplexing approach in FDD massive MIMO

A. Maatouk, S. E. Hajri, M. Assaad, H. Sari

IEEE Transactions on Signal Processing, Vol. 67(4), 1006-1021, 2018

Stay Longer at the Network's Edge: A Novel Proactive Caching Policy through Sojourn Time

J. Denis, A. Maatouk, S. E. Hajri, M. Assaad

IEEE Global Communications Conference (GLOBECOM), 1-6, 2018

A simple NOMA scheme with optimum detection

E. Caliskan, A. Maatouk, M. Koca, M. Assaad, G. Gui, H. Sari

IEEE Global Communications Conference (GLOBECOM), 1-6, 2018

Graph theory based approach to users grouping and downlink scheduling in FDD massive MIMO

A. Maatouk, S. E. Hajri, M. Assaad, H. Sari, S. Sezginer

IEEE International Conference on Communications (ICC), 1-7, 2018

On the foundation of NOMA and its application to 5G cellular networks

H. Sari, A. Maatouk, E. Caliskan, M. Assaad, M. Koca, G. Gui

IEEE Wireless Communications and Networking Conference (WCNC), 1-6, 2018

Frequency-domain NOMA with two sets of orthogonal signal waveforms

A. Maatouk, E. Çalışkan, M. Koca, M. Assaad, G. Gui, H. Sari

IEEE Communications Letters, Vol. 22(5), 906-909, 2018

Invited Talks

Global System for Mobile Communications Association | London, UK | September 2025

Multi-Modal Data Integration for Telecom AI Models

Karlsruhe Institute of Technology | Karlsruhe, Germany | July 2023

Goal-Oriented Semantics Communications: The Role of Data Freshness

Massachusetts Institute of Technology | Cambridge, MA | February 2023

Large Scale Optimization of Cellular Networks: The Role of Machine Learning

Yale University | New Haven, CT | February 2023

Large Scale Optimization of Cellular Networks: The Role of Machine Learning

George Washington University | Washington DC | November 2022
Semantics-Empowered Communications: A New Paradigm for Networked Systems

Middle East Technical University | Ankara, Turkey | April 2021
Toward a Semantics-Empowered Communication Paradigm

École Nationale Supérieure de l'Électronique | Cergy, France | March 2021
Semantics of Data: Age of Information and Beyond

Auburn University | Auburn, AL | March 2020
The Age of Incorrect Information: A New Perspective on Status Update Systems

Tutorials

IEEE Global Communications Conference (GLOBECOM) | Madrid, Spain | December 2021
Age of Information: A Paradigm Shift Towards Semantic Properties in Data Science

IEEE International Conference on Communications (ICC) | Montreal, Canada | June 2021
Semantics in Communications: The Example of Information Freshness

Academic Services

AI/ML Community

- Reviewer for NeurIPS, ICML, ICLR, and KDD

Information Theory Community

- TPC member for IEEE ISIT and IEEE INFOCOM Workshop on Age and Semantics of Information
- Reviewer for IEEE ISIT, IEEE ITW, IEEE Transactions on Information Theory, and IEEE Journal on Selected Areas in Information Theory

Signal Processing Community

- Organizer of the Age of Information symposium at the 57th Annual Asilomar Conference
- Reviewer for the Asilomar Conference and IEEE Transactions on Signal Processing

Networking Community

- TPC member for netAIsys @ ACM MobiSys, IEEE ICC Workshop on The Impact of LLMs on 6G Networks, and IEEE GLOBECOM IMMLLM6G Workshop
- Reviewer for IEEE Infocom, IEEE/ACM Transactions on Networking, and IEEE Journal on Selected Areas in Communications

Mentorship & Supervision

Yale University

- **PhD Students:** Hiren Madhu, Trey Crump, Andreas Varvarigos, Ioannis Panitsas, Eftychia Makri
- **Undergraduate Students:** Neil He, Rishabh Anand, Austin Feng, Rawad Kansoh

George Washington University

- **PhD Students:** Egemen Erbayat, Peng Zhou

University of Maryland, College Park

- **PhD Students:** Yutao Chen, Zahra Zare

Huawei Technologies

- **Interns:** Mert Unsal, Andrei-Laurentiu Bornea, Pedro Luiz Silva, Shi Biao, Henrique Aguiar

Languages

English: Fluent

French: Native

Arabic: Native

Spanish: Elementary